

 OR-AS Operations Research Applications and Solutions	Case Name: Apartment Building (5)	Sector	Construction (residential building)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Matilde Casado and Margarida Antunes		One of nine std. scenarios	
Date	2015		User defined distributions	
File Name	C2015-04 Apartment Building (5)	Project Control	Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

The project includes roofing, ventilation, insulation, electrical, plumbing, flooring, carpentry, kitchen installation, and finishing works, ensuring a fully equipped and comfortable residential complex.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	56	Serial/Parallel (SP)	27%
Planned Duration (PD)	160 days*	Activity Distribution (AD)	37%
Budget At Completion (BAC)	€ 2,750,938	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	57%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

Cost sensitivity				Time sensitivity			
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	11.4	11.9	-2.6	CI	28.6	45.6	-1.0
CRI-rho	11.1	11.6	-2.6	SI	33.6	42.4	-1.0
CRI-tau	6.7	7.5	-3.0	SSI	6.3	12.2	-3.2
Resource sensitivity				CRI-r	13.5	12.0	-2.8
	avg [%]	std dev [%]	skew [-]	CRI-rho	13.0	11.8	-2.9
CRI-r	N/A	N/A	N/A	CRI-tau	8.7	8.2	-3.5
CRI-rho	N/A	N/A	N/A				
CRI-tau	N/A	N/A	N/A				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	5.9	-3.6	1	2.2	0.1
PV - SPI	10.4	6.5	CPI	3.2	0.5
PV - SCI	12.8	7.0	SPI	8.6	8.4
ED - 1	10.2	-9.9	SPI(t)	12.3	12.2
ED - SPI	10.4	6.5	SCI	8.8	8.6
ED - SCI	11.1	6.8	SCI(t)	12.4	12.3
ES - 1	6.6	1.4	0.8 CPI + 0.2 SPI	3.9	2.8
ES - SPI(t)	22.1	21.8	0.8 CPI + 0.2 SPI(t)	5.0	4.3
ES - SCI(t)	22.1	21.9			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 8 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-13942.7	-289149.9	-255.8	1.0	0.9	0.8	1.0
std dev	81222.3	349525.1	246.4	0	0.1	0.2	0.0
final	160141.8	0	-440	1.1	1.0	0.7	1.0

2.3.3.2. Time forecasting

PD	160 days	Real Duration	215 days	Late	34.4%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	161.9	2.0	3.1	3.1
PV - SPI	162.5	2.1	3.1	3.1
PV - SCI	163.1	2.0	3.0	3.0
ED - 1	163.9	1.7	3.0	3.0
ED - SPI	164.6	1.6	2.9	2.9
ED - SCI	165.4	1.6	2.9	2.9
ES - 1	166.1	1.7	2.8	2.8
ES - SPI(t)	166.9	2.0	2.8	2.8
ES - SCI(t)	167.5	2.1	2.8	2.8

2.3.3.3. Cost forecasting

BAC	€ 2,750,938	Real Cost	2 590 796.5 €	Under Budget	-5.8%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	2764881.0	81222.3	0.8	-0.8
CPI	2781261.9	96721.0	0.9	-0.9
SPI	2913286.3	266503.2	1.6	-1.6
SPI(t)	3006613.0	422168.6	2.0	-2.0
SCI	2935332.0	291661.0	1.7	-1.7
SCI(t)	3033044.6	453867.3	2.1	-2.1
0.8 CPI + 0.2 SPI	2801647.8	117792.0	1.0	-1.0
0.8 CPI + 0.2 SPI(t)	2809667.8	132887.5	1.1	-1.1